

LIMBURG REGL, Belgium

WMO: **064700**

Lat: **50.792N** Lon: **5.202E** Elev: **75** StdP: **100.43** Time Zone: **1.00 (EUC)** Period: **82-92** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	-10.5	-7.4	-13.3	1.2	-9.7	-10.2	1.6	-6.3	14.0	8.8	12.8	6.9	3.2	50	0.570	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	10.3	28.3	20.4	26.7	19.7	24.9	18.7	21.3	26.9	20.3	25.6	19.3	23.9	3.3	60	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	19.2	14.1	24.4	18.2	13.3	23.1	17.3	12.5	21.8	62.1	27.4	58.5	25.9	55.1	23.9	27.3	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	11.1	9.6	8.4	DB	-12.4	32.2	6.4	2.3	-17.0	33.8	-20.7	35.1	-24.3	36.4	-29.0	38.1	(4)
(5)				WB	-12.7	23.5	6.3	1.7	-17.2	24.7	-20.9	25.7	-24.4	26.7	-29.0	27.9	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		9.7	2.2	2.1	5.8	8.2	12.5	15.0	17.6	17.2	14.2	11.3	6.0	4.2	(6)
(7)		DBStd		6.56	5.14	4.85	3.42	3.31	3.65	3.39	3.12	2.75	2.99	3.28	3.93	3.76	(7)
(8)		HDD10.0		1027	243	223	135	72	16	2	0	0	2	24	128	182	(8)
(9)		HDD18.3		3204	501	456	388	303	184	111	53	55	127	220	369	438	(9)
(10)		CDD10.0		934	1	1	6	19	92	151	237	224	129	63	9	2	(10)
(11)		CDD18.3		70	0	0	0	0	2	11	31	21	4	1	0	0	(11)
(12)		CDH23.3		736	0	0	0	0	47	110	310	222	43	4	0	0	(12)
(13)		CDH26.7		156	0	0	0	0	5	13	78	55	4	0	0	0	(13)
(14)	Wind	WSAvg		3.7	4.9	4.3	4.3	3.7	3.2	3.0	2.9	2.9	3.2	3.7	3.9	4.3	(14)
(15)	Precipitation	PrecAvg		784	66	59	57	49	61	74	76	81	59	64	65	73	(15)
(16)		PrecMax		1030	124	142	135	116	140	168	228	217	174	174	110	154	(16)
(17)		PrecMin		600	3	5	2	4	22	14	31	16	15	30	6	19	(17)
(18)		PrecStd		116	30	31	28	25	29	34	41	50	38	28	25	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		12.0	14.3	18.1	21.9	26.8	28.3	31.3	31.2	27.3	23.4	16.2	13.3	(19)
(20)			MCWB		10.8	10.7	12.5	13.9	17.6	20.1	22.2	20.8	20.4	17.9	12.2	11.1	(20)
(21)		2%	DB		10.5	11.9	14.9	19.1	24.5	26.4	28.8	28.0	24.3	20.0	14.3	12.0	(21)
(22)			MCWB		8.9	9.7	10.9	12.6	16.6	19.2	21.2	20.0	18.8	15.7	11.7	10.2	(22)
(23)		5%	DB		9.4	9.9	12.4	16.4	22.5	24.3	26.9	25.7	21.8	17.8	12.9	10.6	(23)
(24)			MCWB		8.2	8.3	9.8	11.0	15.8	18.0	20.3	19.4	17.3	14.4	11.0	9.2	(24)
(25)		10%	DB		8.4	8.4	10.7	14.5	20.1	21.9	24.7	23.6	19.6	16.0	11.6	9.4	(25)
(26)			MCWB		7.3	6.9	8.5	10.1	14.8	17.1	19.1	18.3	16.2	13.6	10.2	8.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		10.9	11.4	13.1	14.9	19.0	21.5	23.2	21.7	20.9	18.0	13.3	11.6	(27)
(28)			MCDB		12.0	13.6	17.0	20.8	25.6	26.6	29.2	27.7	25.9	22.4	15.0	12.5	(28)
(29)		2%	WB		9.4	9.8	11.4	13.2	17.5	20.1	21.9	20.6	19.3	16.2	12.2	10.4	(29)
(30)			MCDB		10.1	11.6	14.2	18.0	23.0	25.0	27.8	26.6	23.8	19.0	13.8	11.6	(30)
(31)		5%	WB		8.4	8.5	10.1	11.8	16.1	18.5	20.7	19.8	17.8	15.0	11.2	9.4	(31)
(32)			MCDB		9.2	9.8	12.0	15.5	21.4	23.1	25.8	25.1	20.7	17.3	12.7	10.4	(32)
(33)		10%	WB		7.2	7.1	9.0	10.6	14.9	17.2	19.4	18.7	16.7	13.9	10.2	8.4	(33)
(34)			MCDB		8.3	8.4	10.4	14.0	19.5	21.2	23.9	23.0	19.1	15.8	11.5	9.3	(34)
(35)	Mean Daily Temperature Range		MDBR		4.8	6.3	7.0	9.0	9.7	9.1	10.3	10.2	8.7	7.6	5.9	4.6	(35)
(36)		5% DB	MCDBR		5.2	8.1	9.3	12.9	14.2	13.2	13.8	14.2	13.0	10.1	7.6	4.9	(36)
(37)			MCWBR		4.8	6.5	6.5	7.6	7.7	7.1	7.5	7.5	7.7	6.3	5.6	4.3	(37)
(38)		5% WB	MCDBR		5.1	7.8	8.3	11.8	12.5	12.2	12.9	13.0	10.9	9.3	6.7	4.8	(38)
(39)			MCWBR		4.9	6.4	6.1	7.2	7.4	7.1	7.4	7.3	6.7	6.2	5.2	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.322	0.345	0.378	0.411	0.421	0.419	0.423	0.415	0.386	0.361	0.336	0.320	(40)
(41)		taud			2.409	2.368	2.282	2.210	2.227	2.249	2.239	2.286	2.354	2.418	2.412	2.376	(41)
(42)		Ebn at Noon			703	780	815	827	836	839	828	815	800	751	676	638	(42)
(43)		Edh at Noon			62	83	108	129	133	131	130	119	100	78	62	56	(43)
(44)	All-Sky Solar Radiation	RadAvg			0.77	1.45	2.60	4.14	4.93	5.36	5.08	4.33	3.25	1.92	0.91	0.61	(44)
(45)		RadStd			0.11	0.28	0.37	0.47	0.47	0.47	0.61	0.44	0.33	0.16	0.10	0.09	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page